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Trends in Neurosciences

Review

Epigenetic and metabolic regulation of developmental timing in neocortex evolution

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The human brain is characterized by impressive cognitive abilities. The neocortex is the seat of higher cognition, and neocortex expansion is a hallmark of human evolution. While developmental programs are similar in different species, the timing of developmental transitions and the capacity of neural progenitor cells (NPCs) to proliferate differ, contributing to the increased production of neurons during human cortical development. Here, we review the epigenetic regulation of developmental transitions during corticogenesis, focusing mostly on humans while building on knowledge from studies in mice. We discuss metabolic-epigenetic interplay as a potential mechanism to integrate extracellular signals into neural chromatin. Moreover, we synthesize current understanding of how epigenetic and metabolic deregulation can cause neurodevelopmental disorders. Finally, we outline how developmental timing can be investigated using brain organoid models.

Highlights

The timing and duration of developmental programs in the neocortex determine final neuron number and neocortex size.

Epigenetic mechanisms regulate the timing of all developmental transitions in the developing neocortex, from neural stem cell expansion to neurogenesis and neuronal maturation, and potentially contribute to the protracted development of the human cortex.

Epigenetic reactions are tightly connected to different metabolic pathways that themselves have been implicated in controlling neocortical development and evolution.

Deregulation of epigenetic and metabolic pathways underlies several neurodevelopmental disorders, highlighting the importance of these pathways for proper brain function.

Metabolic-epigenetic interplay emerges as a novel direction for studying the timing of cortical development, and brain organoids offer exciting new experimental approaches.

Timing of developmental programs

The development of the **neocortex** (see Glossary) involves a series of biological programs, starting with neural stem cell and NPC expansion, followed by cell fate commitment and, ultimately, neuronal maturation. While the sequence of these developmental programs is largely conserved across mammals, the onset, duration, speed, and termination of the programs differ between species [1–5]. In evolutionary biology, the differences in timing are referred to as **heterochrony**, and human cortical development is particularly protracted. To refer to this delay in human development, the term **bradychrony** was recently introduced [4]. Differences in developmental timing impact on developmental outcomes including neuron number and neocortex size.

The radial unit hypothesis postulates that the addition of radial units leads to cortical expansion [6], and differences in the duration of the initial phase of neuroepithelial expansion therefore affect the size of the neocortex. The transition of **neuroepithelial cells** into **apical radial glia** was found to be particularly protracted in humans compared with other great apes [7]. This protracted transition is associated with differences in the size of cerebral organoids derived from induced pluripotent stem cells (iPSCs) from three great ape species. In addition to the size of the founder stem cell pool, the length of the neurogenic period contributes to differences in neuron number between species [8]. This is supported by comparison of mouse strains with different lengths of the neurogenic period, which showed increased production of upper-layer neurons in long-gestation strains, underscoring that a longer neurogenic period contributes to neocortical expansion [9]. By contrast, a premature shift from NPC proliferation to differentiation is linked to microcephaly, a neurodevelopmental disorder characterized by a critical decrease in brain size [10,11].

These examples highlight the importance of developmental timing for neocortex development, neurodevelopmental disorders, and neocortex evolution. A series of comprehensive recent

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reviews have discussed different aspects of timing in development and neurogenesis, such as species-specific biochemical reaction rates, mitochondria, and tempo [12–18]. We focus here on the **epigenetic** regulation of developmental transitions during corticogenesis and synthesize how epigenetic deregulation causes neurodevelopmental disorders. We review the role of metabolism in the proliferative capacity of NPCs in the context of neocortex evolution. Finally, we discuss the interplay between metabolic and epigenetic regulation, and outline future research directions including the application of brain organoids to investigate the developmental timing of corticogenesis.

Epigenetic regulation of developmental transitions

Cellular identity is the result of the expression of specific sets of genes. During neocortex development, defined spatial and temporal gene expression programs control NPC expansion, deep- and upper-layer neurogenesis, gliogenesis, and neuronal maturation, and these programs are determined by a combination of transcription and epigenetic factors [19]. Histone methylation has emerged as an important epigenetic mechanism for the initiation and maintenance of cellular identity in the developing neocortex and the maturation of neurons [19–22]. More recently identified post-translational modifications of histones (Table 1), such as histone lactylation, serotonylation, and crotonylation [23–25] are also abundant in the brain, but their role in regulating developmental timing has so far been little explored (see Outstanding questions).

Exit from pluripotency

Epigenetic modifications, such as the histone modifications mediated by Polycomb and Trithorax group complexes, play a key role in cell fate specification and maintenance during embryonic stem cell differentiation and early development, as revealed initially in studies in mice [26]. During early human cerebral organoid development, it was recently shown that the Polycomb repressive complex 2 (PRC2), which mediates H3K27me3, is important during the exit from pluripotency and during neuroepithelial differentiation [27]. This is in line with the accumulation of repressive H3K27me3 domains at genes encoding developmental regulators, the remodeling of chromatin interactions, and the silencing of alternative lineage fates in both mouse and human neuroepithelial cells (Figure 1) [27–29].

Neuroepithelial cell to apical radial glia transition

The transition of neuroepithelial cells to apical radial glia is marked by the onset of neurogenesis and is accompanied by changes in cell junction components and a switch from symmetric proliferative to asymmetric cell divisions [30,31]. Comparative studies of early gorilla, chimpanzee, and human cerebral organoids have provided important insights into differences in the neuroepithelial cell to apical radial glia transition in great apes [7]. The transition was found to be protracted in human, despite comparable developmental molecular trajectories. What differed was the temporal dynamics of the expression of morphogenesis-related genes, most prominently *ZEB2* that encodes a transcriptional regulator of the epithelial–mesenchymal transition. Several human accelerated regions have been linked to *ZEB2* [32]. How the transcriptional programs are regulated during the neuroepithelial cell to apical radial glia transition is currently incompletely understood. It is worth noting that in mouse there is a major rearrangement of H3K27me3 from neuroepithelial cells to apical radial glia, and several tight junction-associated genes acquire a more repressive chromatin configuration, whereas radial glia-associated genes convert to an active chromatin state [28].

Proliferation versus differentiation

During the neurogenic phase, apical radial glia give rise to more committed progeny, mainly basal progenitor cells, and more rarely directly to neurons [30]. Basal progenitor cells, specifically basal

Glossary

Apical radial glia: neural progenitor cells with bipolar epithelial morphology, characterized by two long radially oriented processes that span from the apical (ventricular) surface to the basal (pial) surface of the developing cortex. Cell bodies of apical radial glia reside in the ventricular zone where they form a pseudostratified epithelium which lines the fluid-filled cavity enclosed by the developing brain.

Bradychrony: delayed development of particular traits that nonetheless reach a mature state ('slowed time'), as opposed to neotony which is associated with immature characteristics.

Epigenetics: a variety of mechanisms, including DNA and histone post-translational modifications, that result in different gene activity states in the context of the same primary DNA sequence, thereby allowing fine-grained regulation of gene transcription and cell fate.

Glycolysis: a metabolic pathway in which cells partially break down glucose via enzymatic reactions that do not require oxygen. Glycolysis is one method through which cells produce ATP. The TCA cycle that can follow glycolysis mediates further breakdown of glucose leading to the generation of more energy.

Heterochrony: a distinct tempo, rate, or duration of a developmental stage or process relative to other species.

Metabolic-epigenetic interplay: the deposition of epigenetic modifications onto chromatin depends on substrates and cofactors provided by cellular metabolism, thereby linking the two processes. Likewise, epigenetic regulation impacts on the expression of metabolic genes.

Neocortex: the mammal-specific neuronal sheet which supports higher-order cognitive functions and typically has six layers that cover most of the forebrain.

Neuroepithelial cells: the founder population of epithelial neural stem cells that line the lumen of the neural tube early in development. Neuroepithelial cells proliferate symmetrically to increase their number and produce a tight pseudostratified epithelium before giving rise to apical radial glia at the onset of neurogenesis.

Oxidative phosphorylation (OXPHOS): a cellular metabolic process occurring within mitochondria intermediate progenitors in mouse and basal radial glia (also known as outer radial glia) in human, generate the majority of cortical neurons. Several epigenetic pathways have been shown to regulate the transition from NPC proliferation towards differentiation (Figure 1). Whereas H3K27me3 represses differentiation [33–35], histone acetylation and crotonylation support neuronal differentiation [36–39]. We focus here on PRC2-mediated H3K27me3 owing to recent novel insights from human organoid models.

that leverages the reduction of oxygen to generate high-energy phosphate bonds in the form of ATP. The process comprises a series of oxidation–reduction reactions involving electron transfer from NADH and $FADH_2$ to oxygen via the enzymatic protein, metal, and lipid complexes of the electron transport chain.

In proliferative apical radial glia, proneural genes, such as *Insm1*, *Eomes*, and *Neurog1/2*, are marked by H3K27me3 [28]. In mice with cortex-specific knockout of PRC2 components, loss of H3K27me3 leads to upregulation of neurogenic gene expression programs, causing a premature shift from apical radial glia self-renewal towards differentiation, resulting in a shortened neurogenic period [33,35,40,41]. Likewise, inhibition of PRC2 during the neurogenic phase in human cortical organoids causes a shift from proliferation to differentiation, resulting in reduced organoid size [42]. Thus, epigenetic regulation is important in controlling the timing of NPC differentiation and the length of the neurogenic period, and both are crucial for determining neocortex size.

In addition to regulating neural genes directly, PRC2 was found to regulate non-cell-autonomous mechanisms that drive NPC fate during mouse and human cortical development, such as the composition of the extracellular niche [40–42]. Indeed, a set of temporally patterned genes was suggested to direct apical radial glia from internally driven to more exteroceptive states that are characterized by increased expression of membrane receptors and cell–cell signaling factors during the later stages of mouse development [35]. Taken together, studies in mouse, and more recently in human organoid models, underscore the importance of epigenetically driven temporal patterning in the neocortex.

Neuronal subtype specification

One distinctive characteristic of the neocortex is the organization of neurons into six layers, where deep-layer neurons are born first, followed by upper-layer neurogenesis [43]. Loss of PRC2 in the mouse neocortex has been linked to the precocious production of normally later-born neuronal subtypes [35] and a reduction in upper-layer neurons [40]. Likewise, PRC1 regulates the timed termination of deep-layer neuron generation [44]. In addition to Polycomb-mediated regulation, the histone methyltransferase DOT1L, that is responsible for

Table 1. Summary of metabolic pathway activities and their involvement in epigenetic modifications in mouse and human NPCs

	Apical progenitors		Basal progenitors				Key metabolites for epigenetics	Involvement in epigenetics	Refs for metabolism	Refs for epigenetics
	Apical radial glia		Basal intermediate progenitors		Basal radial glia					
	Mouse	Human	Mouse	Human	Mouse	Human				
Glycolysis	High	–	Low	–[a]	–	–	Acetyl-CoA, lactate	Acetylation, lactylation	[83]	[74,122,123]
Glutaminolysis	High	High	Low	–	–	High	αKG	Demethylation	[90–92]	[116–118]
β-Oxidation	High	–	Low	–	–	–	Acetyl-CoA	Acetylation	[24,87]	[74]
Fatty acid synthesis	Low	High	Low	–	–	–	N/A	N/A	[134]	
Cholesterol metabolism	High	–	Low	–	–	–	N/A	N/A	[135,136]	
1C metabolism	High	–	Low	–	–	–	SAM	Methylation	[115]	[109]

[a]Not determined.

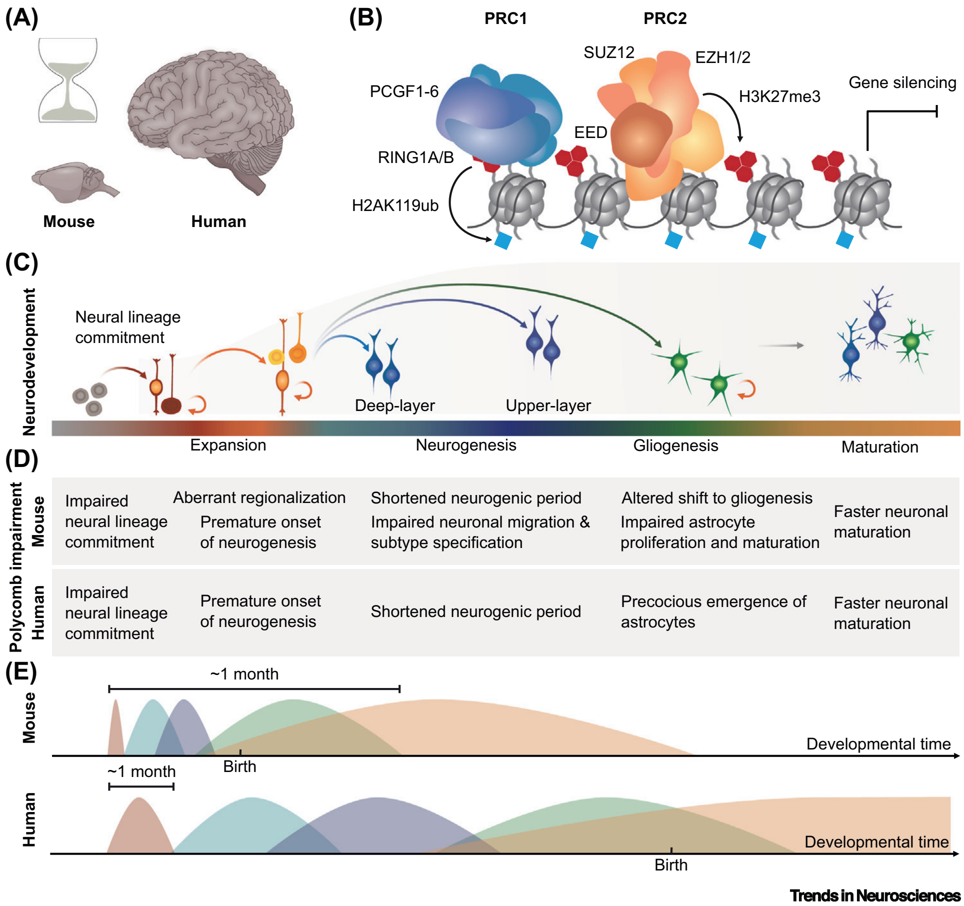

Figure 1. Epigenetic regulation of developmental timing during mouse and human neocortex development. (A) The neocortexes of mouse and human differ in various respects, including developmental timing. Note that the brain illustrations are not to scale. (B) Epigenetic regulation has emerged as an important mechanism to control developmental timing. In particular, Polycomb repressive complex 1 (PRC1) and PRC2, which deposit repressive histone marks – monoubiquitination of lysine 119 of histone H2A (H2AK119ub1) and trimethylation of lysine 27 of histone H3 (H3K27me3), respectively – regulate developmental transitions in the developing neocortex. (C) Key stages of cortical development include early neural lineage commitment, neural progenitor cell (NPC) expansion, neurogenesis, gliogenesis, and neuronal and glial maturation, which continue until after birth. (D) Impaired Polycomb activity affects mouse (top) and human (bottom) neocortex development (see main text for references and details). (E) Human cortical development (bottom) is highly protracted compared with mouse (top), and recent evidence suggests that an epigenetic barrier may contribute to this difference in timing [51]. Developmental stages are colored according to the scheme in panel (C).

H3K79me, balances the transcriptional programs necessary for neuronal layer identity in the developing cortex [45]. These studies highlight the importance of epigenetic regulation by H3K27me3, H2AK119ub1, and H3K79me in the timing of neuronal subtype specification in the mouse (Figure 1).

The sequence of neuronal subtype specification is conserved across species, but quantitative aspects diverge. Compared with other species, the human neocortex displays a higher number of upper-layer pyramidal neurons, which could contribute to higher levels of connectivity between cortical areas [46]. How the timing of human neuronal subtype generation is regulated and whether epigenetic mechanisms contribute to increased production of upper-layer neurons in human are important directions for continued research.

Neurogenic to gliogenic switch

Towards the end of embryonic development in the mouse, the remaining progenitor cells switch from neurogenesis to gliogenesis. Epigenetic mechanisms, particularly Polycomb regulation, modulate the timing of this switch [33,47–49], thereby affecting neuron number and neocortex size. The switch from neurogenesis to astrogenesis also occurs in cultured NPCs without additional stimuli, suggesting that this is a cell-intrinsic transition [50]. In addition to affecting the switch, loss of PRC2 causes defects in immature astrocyte proliferation and reduced morphological complexity of mature astrocytes [40].

In human cerebral organoids, inhibition of PRC2 has also been shown to cause precocious emergence of astrocytes [51], indicating that epigenetic regulation contributes to the neurogenic to astrogenic switch in the human developing neocortex. Recently, µBrain, a comprehensive resource of the human fetal brain, revealed that the developmental timing of the neurogenic to astrogenic switch shows regional variation across the developing neocortex and is linked to differential rates of neocortical expansion [52]. In gyrencephalic species, basal radial glia are centrally involved in gliogenesis and contribute to neocortical expansion and the development of cortical folds [53–55].

So far, human organoid models have fallen behind in their capacity to model the expansion of the outer subventricular zone and of gliogenesis. Recent organoid protocol improvements involving the addition of leukemia inhibitory factor (LIF) [56,57] might enable future functional studies of the neurogenic to gliogenic switch in human cortical development. This would also facilitate the investigations of the role of other epigenetic modifications in the neurogenic-gliogenic switch in human.

Neuronal maturation

Neuronal maturation is a highly protracted process in human, and lasts for years compared, for instance, with only weeks in mouse [4,58–60]. It was recently shown that an epigenetic barrier controls the timing of neuron maturation by maintaining maturation genes in a poised state and by limiting the expression of competing epigenetic factors that promote maturation [21,51]. Transient inhibition of histone methyltransferases mediating H3K9, H3K27, and H3K79 methylation at the NPC stage primed human neurons to rapidly express maturation programs. Moreover, a bipartite signature, composed of H3K27me3 and H3K27ac, was found to regulate the transcriptional response to stimuli in sensory neurons in the mouse [61].

Taken together, epigenetic mechanisms regulate the timing of virtually all developmental transitions in the developing neocortex, from neural stem cell expansion to neurogenesis and neuronal maturation. Among epigenetic mechanisms, Polycomb complexes have emerged as key regulators of developmental timing in the mouse developing neocortex, and more recent work suggests that Polycomb-mediated regulation may also contribute to protracted brain development in human (Figure 1). Cerebral and cortical organoid models, combined with large-scale epigenomic approaches [27,42,62–64], represent promising tools to dissect further the contribution of other epigenetic modifications in human neocortex development and to gain a better understanding of other primate species.

Neurodevelopmental disorders caused by mutations in epigenetic enzymes

The importance of epigenetic regulation in human cortical development also becomes clear from the many Mendelian disorders of the epigenetic machinery that display neurological dysfunction and intellectual disability [20,65–67]. Several of these developmental disorders affect brain size, resulting in microcephaly (decreased size) or macrocephaly (increased size) (Figure 2).

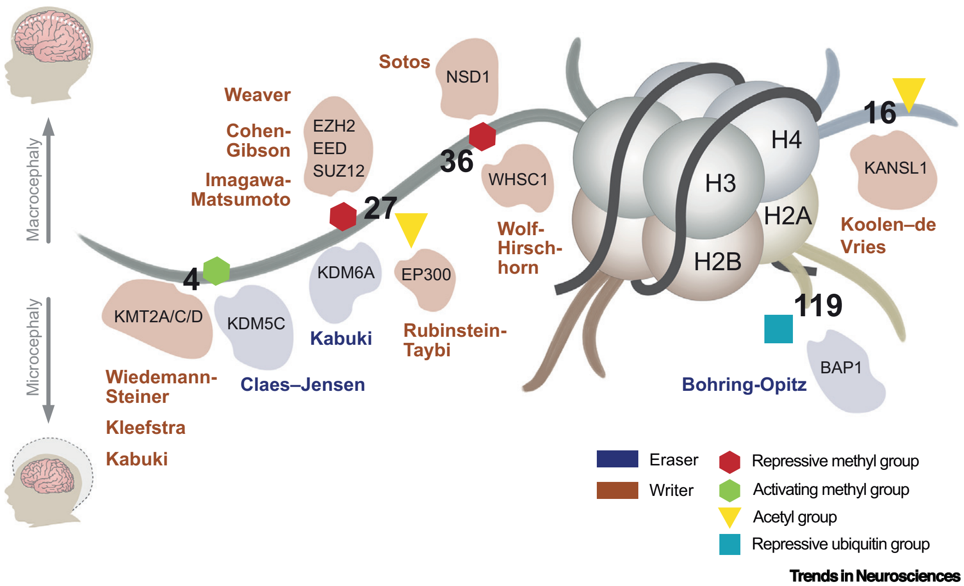

Figure 2. Developmental disorders affecting brain size caused by mutations in histone modifiers. Mutations in various histone modifiers that deposit or erase histone methylation, acetylation, or ubiquitination marks cause developmental disorders linked to impaired brain development. Disease-associated alterations in the epigenetic machinery induce brain phenotypes including microcephaly (decreased size) or macrocephaly (increased size). These disorders highlight the importance of activating and repressive histone modifications in controlling human cortical development.

Heterozygous mutations in any of the core components of PRC2, that are all required for H3K27me3, are linked to macrocephaly and intellectual disability associated with Weaver syndrome (*EZH2*), Imagawa–Matsumoto syndrome (*SUZ12*), and Cohen–Gibson syndrome (*EED*) [68–70]. By contrast, mutations in the H3K27 demethylase gene *KDM6A* or the H3K4 histone methyltransferase gene *KDM2D* cause Kabuki syndrome that is often associated with microcephaly [71,72]. Given the importance of H3K27me3 in regulating the timing of developmental programs in the neocortex, it is unsurprising that alterations in the balance of active and repressive histone modifications result in altered brain development. Intriguingly, however, loss of function of PRC2 during the neurogenic period results in reduced cortical size in both mouse and human models [33,35,40,42] and is thus opposite to the macrocephaly phenotype described for individuals with mutations in genes encoding PRC2 components. Most Weaver syndrome-associated mutations in *EZH2* are predicted to be loss-of-function mutations and may even have dominant negative effects, leading to strongly decreased H3K27me3 levels, as well as transcriptional upregulation of sensitive Polycomb target genes [73]. This suggests that epigenetic regulation of NPC fate alone is likely insufficient to explain the macrocephaly phenotype, and that additional mechanisms, such as cell-extrinsic factors, may contribute to the etiology of this developmental disorder, but this requires further investigation in models carrying patient mutations.

Moreover, brain size is also affected by mutations in several other epigenetic factors (Figure 2) that are linked not only to histone methylation but also to histone acetylation. Histone acetylation facilitates gene expression by increasing chromatin accessibility and promoting transcriptional activation [74]. In the developing neocortex, histone acetylation is important for the proper differentiation of NPCs into neurons, and defects in the histone acetyltransferase CBP in humans are associated with Rubinstein–Taybi syndrome [39]. Of note, several layers of epigenetic regulation may cooperate to safeguard neocortex development, as exemplified by DOT1L and PRC2 interactions [21,75]. Thus, understanding the crosstalk of epigenetic pathways in neurodevelopmental disorders will be an important avenue of future research, especially in the context of developmental timing.

How can such knowledge inform novel therapeutic approaches? While micro- and macrocephaly are typically diagnosed after birth when the generation of embryonic neurons is complete, encouraging data from mouse models of Kabuki syndrome suggest that counteracting the epigenetic deregulation by inhibiting opposing epigenetic pathways can rescue adult neurogenesis, leading to improved memory and hormone regulation [76,77]. Similar approaches may be explored for some forms of autism spectrum disorder where epigenome alterations are central to its etiology [78].

Given the differences in developmental timing and in the abundance of different NPC types in human compared with mouse, human disease models can add additional valuable insights. However, it will be essential to apply rigorous quality criteria to the organoid models with respect to spatial organization, metabolic state, and cell survival to obtain meaningful results [79]. Current benchmarking efforts against human fetal tissue are aiding the selection of organoid protocols [80]. Reassuringly, the epigenomes of human neural organoids and primary fetal tissue are highly similar [27,42,81,82]. Future improvements of organoid models with respect to the abundance of basal radial glia and the generation and arrangement of neuronal subtypes, astrocytes, and oligodendrocytes will further enhance the investigation of human-specific aspects of neurodevelopmental disorders.

Metabolic regulation of cortical development

In addition to epigenetic regulation, cell metabolism controls developmental programs both directly and indirectly. Examples of such direct and indirect regulations of cortical development involve providing essential metabolites for DNA synthesis, a prerequisite for cell proliferation, and for epigenetic modifications which regulate gene expression.

Energy production

Metabolism regulates NPC behavior, especially their proliferation and differentiation [16,83–87]. Pioneering work showed that distinct NPC types exhibit different metabolic activities in the embryonic mouse neocortex (Table 1) [83]. There are two major pathways that produce ATP: **glycolysis** and **oxidative phosphorylation (OXPHOS)**. Due to the hypoxic nature of the ventricular zone in the developing neocortex, which has few blood vessels, apical radial glia predominantly utilize glycolysis rather than OXPHOS to produce ATP. By contrast, in the subventricular zone, which has higher angiogenic activity than the ventricular zone, basal intermediate progenitors produce more ATP by OXPHOS [83,85]. In addition, neurons exhibit higher OXPHOS levels than NPCs (Figure 3). This shift in metabolism, from glycolysis to OXPHOS, suggests that cell metabolism plays a key role in the timing of NPC proliferation and differentiation. In the embryonic mouse neocortex, proliferative NPCs exhibit higher anaerobic glycolysis than neurogenic NPCs [88]. In addition, a sustained high level of lactate-producing glycolysis suppresses neuronal differentiation of NPCs, indicating that OXPHOS is particularly important for neurogenesis [89].

Amino acid metabolism

Recent studies have highlighted the importance of glutaminolysis, a metabolic pathway that converts glutamine to glutamate and then to α-ketoglutarate (αKG), in human NPC expansion which is implicated in the evolutionary enlargement of the human neocortex [84,90–92]. Glutaminolysis supports the expansion of apical radial glia early in neocortical development, and the expansion of basal progenitors, especially basal radial glia, at later stages (Table 1). MCPH1, a microcephaly-associated protein that is localized on mitochondria of apical and basal radial glia, has been shown to promote glutaminolysis by elevating the Ca^{2+} concentration in the mitochondrial matrix [90]. In addition, a human-specific mitochondrial matrix protein, ARHGAP11B, also increases the

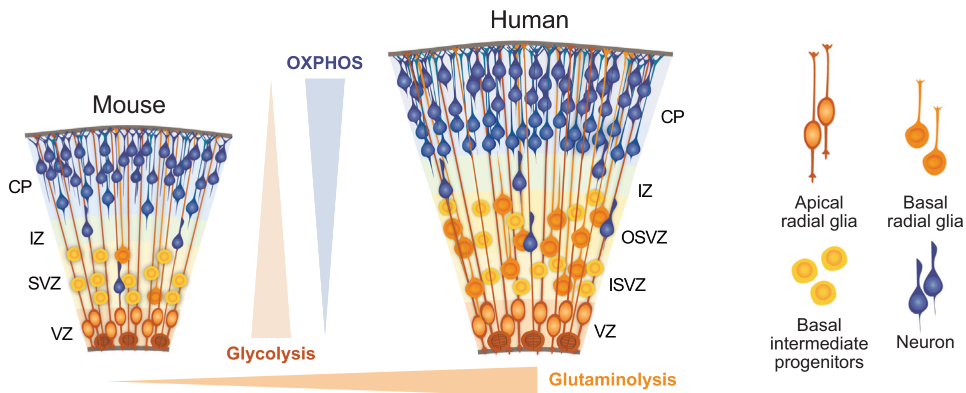

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Figure 3. Cell type and interspecies comparisons of cell metabolism in cortical development. Glycolysis is prominent in apical radial glia in the ventricular zone (VZ) and is reduced during neuronal differentiation. Reciprocally, oxidative phosphorylation (OXPHOS) is highest in the neurons located in the cortical plate (CP) and intermediate zone (IZ), followed by basal progenitors in the subventricular zone (SVZ) in mice and outer and inner SVZ (OSVZ and ISVZ, respectively) in humans. Apical radial glia in the VZ have the lowest level of OXPHOS. Glutaminolysis is higher in human than in mouse neural progenitor cells.

Ca^{2+} concentration in the mitochondrial matrix, thereby inducing glutaminolysis [92]. This effect on glutaminolysis is further promoted by the presence of an ape-specific mitochondrial enzyme, GLUD2 (glutamate dehydrogenase 2), which catalyzes the conversion of glutamate to αKG [10] (Figure 4A). A part of αKG goes through the tricarboxylic acid (TCA) cycle and ends up as

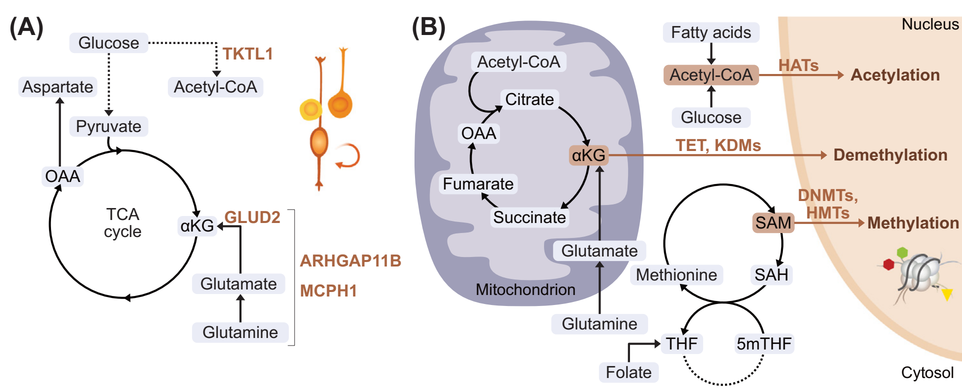

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Figure 4. Metabolic regulation of cortical development and its interplay with epigenetic processes. (A) Changes in genes encoding various metabolic proteins have been implicated in driving human brain development and evolution. Among them, the human-specific protein ARHGAP11B, the ape-specific protein GLUD2, and the microcephaly-associated protein MCPH1 promote glutaminolysis [90–92]. Glutaminolysis has been linked to neural progenitor cell (NPC) expansion which is key for the evolutionary expansion of the neocortex. Moreover, the production of acetyl-CoA is increased in highly proliferating human NPCs, such as basal radial glia. In line with this, the TKTL1 enzyme in modern humans exhibits higher activity and produces more acetyl-CoA than the ancestral versions found in Neanderthals or primates, and thus promotes the higher proliferative capacity of human basal radial glia. (B) The deposition and removal of epigenetic marks, including DNA and histone modifications, requires substrates provided by a plethora of metabolic pathways. Histone acetylation is mediated by histone acetyltransferases (HATs) using acetyl-CoA as a donor of acetyl groups. Acetyl-CoA is produced from glucose catabolism or fatty acid oxidation. Ten-eleven translocation (TET) methylcytosine dioxygenases and lysine-specific histone demethylases (KDMs) are two classes of enzymes responsible for DNA and histone demethylation, respectively. Because α-ketoglutarate (αKG) is the essential substrate for demethylation by these enzymes, αKG levels are critical for the methylation/demethylation balance. DNA and histone methylation is mediated by DNA methyltransferases (DNMTs) and histone methyltransferases (HMTs) which require *S*-adenosylmethionine (SAM), an intermediate metabolite in the methionine cycle, as a methyl donor. Abbreviations: 5mTHF, 5-methyltetrahydrofolate; OAA, oxaloacetic acid; SAH, S-adenosylhomocysteine; THF, tetrahydrofolate.

oxaloacetate, which is in turn converted to aspartate [91,93]. Aspartate is a nitrogen donor for nucleotide *de novo* synthesis, and thus serves as a rate-limiting metabolite for cell proliferation [93]. Consuming aspartate to produce asparagine by asparagine synthetase induces neuronal differentiation of NPCs [75].

The activity of glutaminolysis decreases during the perinatal period owing to increased expression of glutamine synthetase, the enzyme that converts glutamate to glutamine. Increased expression of glutamine synthetase is observed in the radial glia [94]. Induced glutamine production from glutamate suppresses their proliferative capacity and turns them into quiescent stem cells for adult neurogenesis [94]. Glutamine synthetase expression is also prominent in astrocytes [95] which take up excess glutamate produced by neurons. Glutamate is then converted into glutamine, which is finally transported back into neurons where glutamine is utilized for glutamate production. Both hyper- and hypoactivation of the glutamate–glutamine cycle, owing to constitutively or partially active glutamine synthetase pathogenic variants, respectively, are thought to be a cause of epilepsy [96,97]. These studies suggest that amino acid metabolism is one of the key determinants that balance NPC proliferation and differentiation, as well as being a critical factor for maintaining amino acid homeostasis in the postnatal brain.

Acetyl-CoA production

Acetyl-CoA is a key metabolite for fatty acid biosynthesis and histone acetylation. The production of acetyl-CoA from glucose has been shown to be upregulated in human basal radial glia [98]. Transketolase-like 1 (TKTL1) is a key enzyme of the pentose phosphate pathway (PPP) and typically contains arginine at amino acid residue 261 in modern humans. However, the TKTL1 found in Neanderthals and primates (referred to as ancestral TKTL1) contains a lysine residue at the same position. Through this amino acid substitution, modern human TKTL1 obtained higher activity to produce acetyl-CoA, which in turn leads to an increase in the proliferative capacity of basal radial glia in human neocortex. Since TKTL1 expression is higher in the frontal lobe of the fetal human neocortex than the occipital lobe, modern human TKTL1 might contribute to the difference in frontal lobe size between modern humans and Neanderthals [98]. These studies further highlight the significance of metabolism and mitochondrial activity in neocortical development and evolution (Figure 4A).

Mitochondria

Mitochondria are organelles that show dynamic changes in their morphology and distribution. The distribution of mitochondria from mother cell to daughter cells needs to be tightly regulated during cell division [16,86,99]. During mitosis of NPCs, mitochondrial fission is induced, presumably for the proper distribution of mitochondria to daughter cells. After cytokinesis, mitochondrial dynamics differ among daughter cells with distinct cell fates. Daughter cells staying in a progenitor state have higher mitochondrial fusion activity and thus have elongated mitochondria. By contrast, daughter cells undergoing neuronal differentiation show lower mitochondrial fusion activity, and their mitochondria are fragmented [100]. Since inhibition of mitochondrial fusion promotes neuronal differentiation of daughter cells, the morphological dynamics of mitochondria seem to play an important role in NPC fate determination. Interestingly, human NPC progenies take longer to develop elongated mitochondria than the progenies of mouse NPCs, leading to slower fate determination. Overall, these findings suggest that the morphological dynamics of mitochondria, and likely their activity, are a determinant of NPC fate decision and affect the timing of neurogenesis [16].

Neurodevelopmental disorders resulting from metabolic dysregulation

The importance of metabolism also becomes clear from studying cortical malformations. *SLC25A19* is a gene linked to Amish lethal microcephaly [101,102]. Mutation of *SLC25A19* leads to a reduced amount of the encoded protein, a mitochondrial thiamine pyrophosphate carrier, which results in insufficient thiamine pyrophosphate for αKG dehydrogenase complex activity in mitochondria, thereby attenuating αKG catabolism. Amish lethal microcephaly highlights the importance of αKG catabolism in normal brain development, which is corroborated by studies indicating the importance of glutaminolysis [84,90,91].

Methionine, an essential amino acid, and folate (vitamin B9) must be provided through the diet. For humans, folate is less available in the daily diet since it is heat-sensitive. Women are recommended to take folic acid before conception and during the first trimester of pregnancy because folic acid, which is converted into folate in the body, significantly reduces the risk of spina bifida, a neural tube closure deficiency. The relevance of DNA and histone methylation in neural tube closure and brain development needs to be studied further [103–106].

Cell metabolism as a regulator of epigenetic modifications

Epigenetic modifications of DNA and histones require specific substrates and cofactors, the cellular supply of which is tightly regulated by cell metabolism. Therefore, changes in cell metabolism can influence the epigenetic status of a cell (Figure 4B) and thus alter cell fate determination.

DNA and histone methylation

S-adenosylmethionine (SAM), a derivative of methionine, is the methyl donor for DNA and histone methylation. DNA methyltransferases DNMT1, DNMT3A, and DNMT3B transfer a methyl group of SAM to cytosine to produce 5-methylcytosine with *S*-adenosylhomocysteine (SAH) as a byproduct [107,108]. SAM is also the methyl donor for histone methylation. Histone methyltransferases transfer a methyl group from SAM to histones and produce methylated histones such as H3K27me3 as well as SAH [109]. Therefore, SAM is the rate-limiting metabolite for DNA and histone methylation. SAM is an intermediate metabolite in the methionine cycle. The methionine and folate cycles are inseparable and constitute the major pathways of one-carbon (1C) metabolism [110,111]. Changes in the activity of 1C metabolism, through alterations in either the methionine cycle or the folate cycle, are known to influence the epigenetic status of a cell [112–114]. Notably, the H3K4me3 level in apical radial glia has been linked to neuronal differentiation [115].

DNA and histone demethylation

DNA and histone methylation can be actively removed by enzymes, thereby facilitating epigenetic regulation during development. DNA cytosine methylation is converted to 5-hydroxymethylcytosine by ten-eleven translocation (TET) methylcytosine dioxygenases, which utilize αKG [116]. Lysine-specific histone demethylases can demethylate lysine residues of histones. Of eight histone demethylase subclasses, KDM1 requires flavin adenine dinucleotide (FAD), whereas other histone demethylases utilize αKG to demethylate histones, indicating that the intracellular level of αKG is a critical factor in the DNA and histone methylation–demethylation balance [117,118]. Therefore, increased production of αKG in human NPCs [84,91] may induce demethylation of DNA and histones, thereby maintaining the high proliferative activity of human NPCs. Indeed, histone methylation is generally lower in NPCs compared with neurons [42]. Interestingly, intracellular αKG is known to be associated with the levels of DNA and histone methylation, and is crucial for maintaining the pluripotency of mouse embryonic stem cells [119].

Histone acetylation

Acetyl-CoA is the donor of acetyl groups for histone lysine acetylation mediated by histone acetyltransferases. Histone deacetylation is regulated by a family of histone deacetylases [120]. Class III histone deacetylases, namely sirtuins, depend on NAD to exert their activity. Therefore, cell metabolism controls the ratio of the oxidized to the reduced form of NAD (NAD^+/NADH ratio) and influences the activity of sirtuins and thus histone deacetylation [121]. One metabolic pathway controlling NAD/NADH metabolism is mitochondrial OXPHOS. Interestingly, inhibition of OXPHOS increases the NAD/NADH ratio and promotes neuronal differentiation of NPCs, which is presumably mediated by the activation of sirtuins and histone deacetylation [100].

Histone lactylation

While histone methylation and acetylation are among the best-studied types of modifications, there are several other histone modifications. Lactylation at H3K18 is known as a mark of active transcription [122]. Cells with higher lactate-producing glycolysis, such as neural crest cells, show a substantial level of histone lactylation [123]. Since histone lactylation is possibly primed by the transcription factors SOX9 and YAP/TEAD, and these factors are important regulators of human NPC stemness [124,125], human NPCs might harbor a higher level of histone lactylation. Lactyl-CoA, which is produced from lactate and acetyl-CoA, is the key metabolite for histone lactylation. Lactyl-CoA is produced in the cytoplasm and then transported into the nucleus. Interestingly, a recent study showed that lactyl-CoA is also produced in the nucleus *in situ* by GTP-specific succinyl-CoA synthetase [126]. In this case, lactate itself needs to be transported into the nucleus.

Taken together, cell metabolism and epigenetics appear to be tightly connected, and their interplay may play a pivotal role in the timing of NPC proliferation and differentiation (Table 1). Elucidating the metabolic and epigenetic interplay in human brain development and evolution is an important future goal for the field.

Concluding remarks and future directions

The regulation of developmental timing is tightly linked to interspecies differences in developmental programs and, in the developing neocortex, directly impacts on neuron number and thus on neocortex size. Epigenetic mechanisms are central to the regulation of developmental gene expression programs, and we argue that histone modifications contribute to regulating the timing of cellular proliferation in the developing neocortex. While there is already strong evidence, from both mouse and human studies, for the involvement of epigenetic factors such as Polycomb complexes in regulating developmental timing, the contribution of other epigenetic pathways requires further exploration. In particular, the epigenetic factors whose mutation results in neurodevelopmental disorders affecting brain size represent strong candidates for future study.

Moreover, both epigenetic mechanisms and metabolism have been linked to the regulation of developmental timing in the developing neocortex, but the **metabolic-epigenetic interplay** remains to be further explored in this context. Cell metabolism regulates the supply of specific substrates and cofactors required for DNA and histone modifications, and changes in cell metabolism therefore influence epigenetic states, which in turn affect neural cell fate. Conversely, epigenetic modifications such as H3K79me regulate the expression of metabolic genes, and epigenetic mechanisms may be involved in metabolic diseases [75,127]. Of note, the importance of the metabolic-epigenetic interplay has been recognized in cancer research, and strategies are emerging for how this interplay can be exploited to develop novel therapeutic approaches [128,129].

Advanced neural organoid models from human and non-human pluripotent stem cells, including from individuals with neurodevelopmental disorders, provide unprecedented opportunities to investigate the functional importance of epigenetic and metabolic regulation of developmental timing (Figure 5). It is essential, however, to monitor metabolic stress [80,130,131] in organoid cultures, an aspect which varies across different organoid protocols [80]. More generally, it is

Outstanding questions

Histone methylation has emerged as being central to the regulation of developmental timing in cortical development. What is the contribution of other epigenetic modifications? Do epigenetic mechanisms contribute to the heterochrony of human brain development?

How does the crosstalk of different epigenetic pathways affect neural progenitor proliferation and differentiation in normal development and in neurodevelopmental disorders?

Epigenetic and metabolic regulation are both important for developmental timing and neocortex evolution. How does the metabolic-epigenetic interplay regulate neural stem cell expansion, neurogenesis, and neuronal maturation? Can insights into this interplay provide novel strategies for therapeutic intervention in neurological disorders?

How does dietary uptake of micronutrients, such as folic acid, affect DNA and histone methylation, and therefore gene regulation, during cortical development?

Are global differences in histone modifications between NPCs and neurons the result of different metabolic regulation? Conversely, do global epigenetic changes in neurodevelopmental disorders alter the expression of metabolic genes?

What lessons can evolution teach us regarding epigenetic and metabolic regulation? Are the same pathways altered in neurodevelopmental disorders? Can the reversibility of epigenetic marks be exploited in treating human diseases?

Brain organoid models offer unprecedented opportunities to explore the metabolic-epigenetic interplay and its functional role in regulating developmental timing. How can organoid culture conditions be further improved to reduce metabolic stress and increase basal radial glia and astrocyte populations?

What can functional investigations of the metabolic-epigenetic interplay teach us about developmental timing?

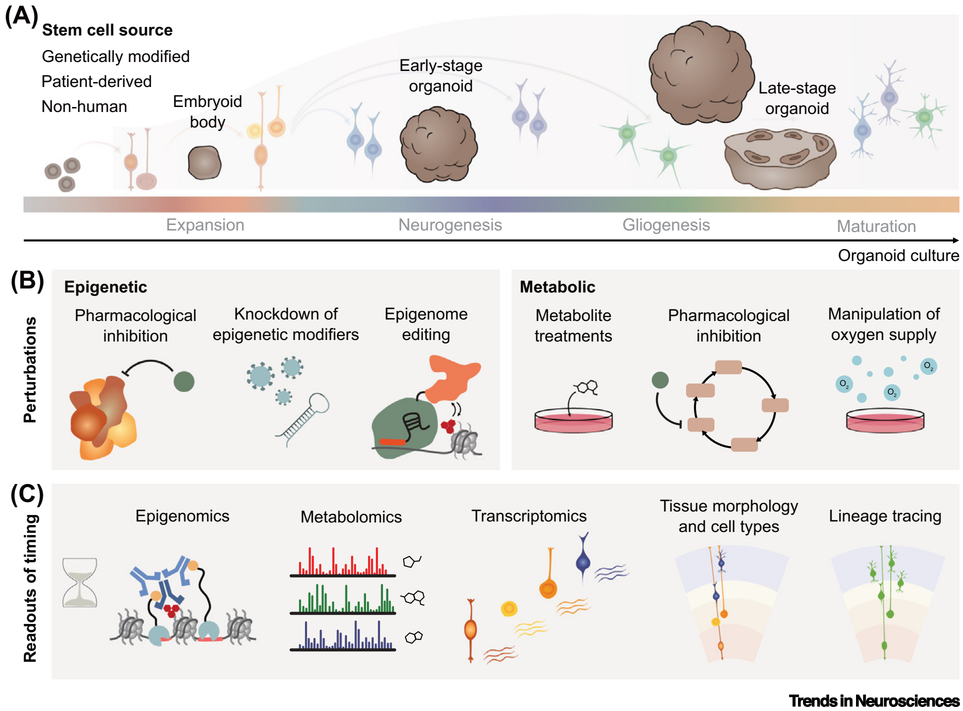

Figure 5. Brain organoids enable the investigation of epigenetic and metabolic regulation of the developmental timing of corticogenesis. (A) Cerebral and cortical organoid models generated from various stem cell sources recapitulate major stages of corticogenesis, including different neural progenitor cell (NPC) populations and tissue morphology across the trajectory of development. Therefore, organoid models present an unprecedented model system for functional investigations of the timing of neocortex development. (B) Functional studies of the metabolic-epigenetic interplay require perturbations of the epigenetic and metabolic machinery, which can be implemented in organoid systems by a range of different approaches. (C) Combining these functional perturbations with readouts of the epigenetic landscape and the metabolic state at cell type level or single-cell resolution will greatly enhance our understanding of metabolic-epigenetic interplay in controlling the timing of corticogenesis.

Epigenetic inhibitors, epigenome editing, and metabolic perturbations can all be applied to brain organoids to address this question.

important, when feasible, to validate findings from *in vitro* models with primary fetal tissue, for example, by using the free-floating tissue culture approach [132], as was previously done in studies of epigenetic and metabolic regulation [42,91,92,98,133]. Cortical organoids, in combination with epigenetic inhibitors, epigenome editing tools, and metabolic perturbations, provide unique opportunities to dissect the metabolic-epigenetic interplay in neocortex development and evolution, as well as in neurodevelopmental disorders. The aspect of developmental timing deserves particular focus because of its impact on neuron number and neocortex size. Shedding light on the metabolic–epigenetic interplay not only promises novel insights into what makes us human but may also highlight promising targets for therapeutic interventions in neurological disorders.

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Declaration of interests

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Grammarly for spelling and grammar checks. After using the tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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